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标题	Carbonylative Synthesis of β - Trifluoromethylated Heterocyclic Ketones through 1,2-Trifluoromethylation and Minisci Carbonylation of Alkenes
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摘要（原文节选）

The present study details a metal-free catalytic Minisci carbonylation reaction, founded upon a radical polarity reversal strategy. The reaction under scrutiny involves the reaction of an electrophilic trifluoromethyl radical with an electron-rich alkene to generate a nucleophilic alkyl radical, which then undergoes a Minisci carbonylation transformation with various electron-deficient N-heteroarenes under a carbon monoxide atmosphere. This method has been demonstrated to be both universal and efficient in the synt ⋯

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Optimization of the Reaction Conditions

Entry	Oxidant	Solv.	Yield 4a (%)
1	TBPA	DMSO	35
2	DTBP	DMSO	N.D.
3	TBHP	DMSO	30
4	TBPA	DMSO	45
5	TBPA	DMF	17
6	TBPA	MeCN	13
7	TBPA	THF	N.D.
8 ,	TBPA	DMSO	50
9 , ,	TBPA	DMSO	60
10 , , ,	TBPA	DMSO	52
11 , , , ,	TBPA	DMSO	N.D.

脚注（逐字）：aReaction conditions: 1a (0.2 mmol), 2a (0.3 mmol), 3a (0.1 mmol), TBPA (3 equiv), TFA (2 equiv), DMSO (1 mL), H2O (0.5 mL), CO (40 bar), rt, 24 h. Isolated yieldsb 1a (0.3 mmol), 2a (0.2 mmol), and 3a (0.1 mmol).cTBPA (3.5 equiv).dTFA (6 equiv).eCO (30 bar).fWithout TBPA. TBPA = tert-butyl peroxyac

正文对应句（逐字）：In accordance with the aforementioned reaction design, the utilization of hex-1-ene (1a), CF3SO2Na (2a), and 4-methylquinoline (3a) as template substrates enabled the employment of tert-butyl peroxyacetate (TBPA) as the oxidant. The reaction was conducted at ambient temperature under a 40 bar CO atm

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